VARĖNA · C. 1902 · KVR 32911

Vilnius Military District Artillery Range Magazines

What the Tsarist army magazines hidden under earth looked like: from today’s earth mound in the forest to a 3D reconstruction



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Photogrammetry, 3D models, visualisations · 2026

Created as part of the project: “Fort-Union – Integration of the Polish–Lithuanian Network of Fortifications for the Development of Representative Tourism Products”

A mound that hides the magazines

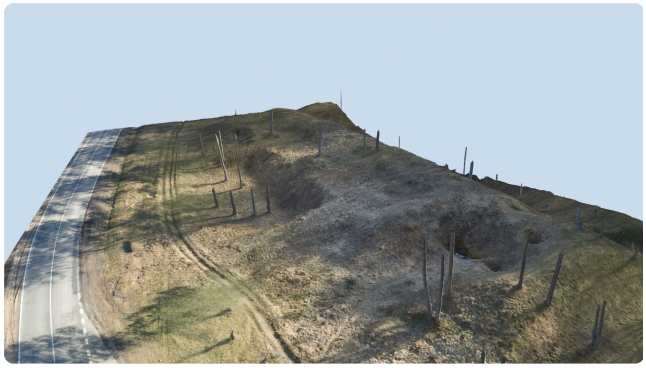
In a pine forest beside the Varėna–Alytus road lies an earth mound almost 90 metres long. Beneath it are the Vilnius Military District Artillery Range Magazines, built around 1902: interconnected red-brick rooms with barrel vaults, used to store pyroxylin (smokeless powder) and other explosives.

In 1877 a large artillery range with a summer camp was planned near Varėna (then Orany). Pyroxylin needed a constant temperature, protection from moisture and maximum safety, so the magazines were built semi-underground: the earth cover damped a possible blast and protected against temperature swings. They were used by the Tsarist Russian and Imperial German armies, were not damaged in the war, and in early 1919 passed in very good condition to the local committee. Documents from 1922 call them an artillery cellar and record rooms of 31 × 2 m, 2.5 m high. The Lithuanian army did not use them: in 1924 they were handed over for dismantling, and the bricks and other materials were given to new settlers. This is why the site looks so altered today; what remained decayed and was overgrown by forest. Today the mound, brick walls, vaults and three semicircular openings in the north-eastern part of the complex can be seen.

~89 m	~4.9 m	3	~1902
mound length	mound height	vaulted chambers	year built



The mound today, drone view. Photo: S. Zaura, 2026.



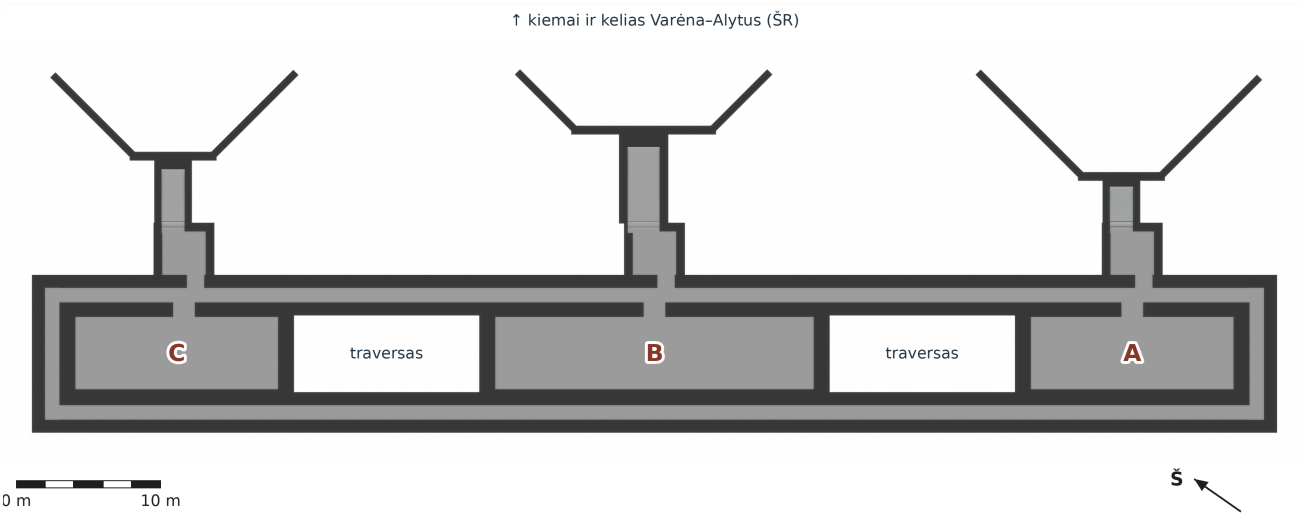
3D model of the current state, created from hundreds of photographs (photogrammetric recording).

> **Safety.** The ruins are dangerous: the collapses are deep and unexploded ordnance may remain in the area. Do not enter.

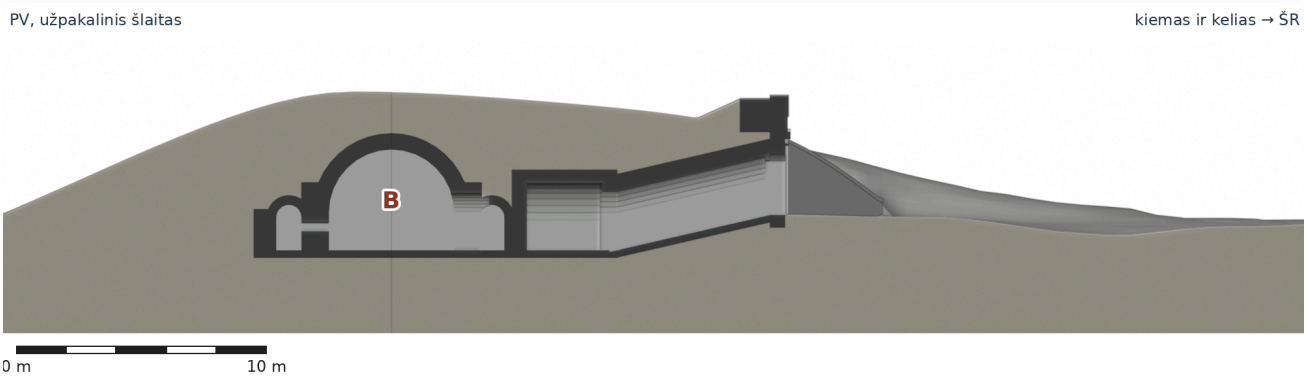
How their appearance was reconstructed

No drawings of the magazines have been found in archives so far, and the interior is filled with debris. The reconstruction therefore combines three kinds of evidence:

- **measured** shape, length and height of the mound and the collapses, from the 3D model of the current state;
- **based on analogues** layout of chambers, gallery and entrances, from period rules for powder magazines and similar surviving fortress stores in Kaunas, Grodno, Warsaw and Modlin;
- **hypothesis** exact dimensions and portal decoration, as reasoned assumptions.



Plan: three chambers (A, B, C) surrounded by a narrow ventilation gallery. Each chamber has its own entrance with a dog-leg passage. (Labels in Lithuanian: traversas = earth traverse; Š = north.)



Cross-section through the main chamber B: the vault under earth cover and the sloping passage to the yard.

Why so complicated?

Everything served safety. Thick load-bearing walls and earth traverses separated the chambers, so an explosion in one would not spread to the others. The dog-leg passage broke the blast wave. Lamps burned in the gallery behind glazed windows, so no open flame entered the chamber. Wagons never drove inside: soldiers carried the crates by hand.

How it may have looked in the early 20th century



Main entrance B. artistic interpretation



All three entrances from above. artistic interpretation



The approach road to the magazines. artistic interpretation

The photorealistic images are renders of the 3D model refined with auxiliary AI image-processing tools. The building follows the reconstruction; people, horses and wagons are illustrative.

Inside and in section



Chamber B with storage racks. artistic interpretation



Ventilation gallery with lamp windows. artistic interpretation



Longitudinal section: the chambers under the earth mound. Render of the 3D model.

Explore the 3D models

Current state, 2026
skfb.ly/plBzL

Reconstruction
skfb.ly/pOASQ

Reconstruction, cut-away
skfb.ly/pOARJ

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